

Bike Buyer Conversion Analysis

An end-to-end Excel project analyzing customer data for a bike retailer, built to answer a simple business question: **who actually buys bikes, and what separates a buyer from a non-buyer?**

What's in this project

- **Raw data** — 1,026 customer records: demographics (age, gender, marital status, education, income, occupation), household info (children, home ownership, cars), commute distance, region, and whether they purchased a bike.
- **Cleaned working sheet** — data validated and enriched with a derived `Age Range` column (Young / Middle Age / Old Buyer) using nested IF logic, so age can be analyzed as a category instead of just a raw number.
- **Pivot tables** — cross-tabs of average income by gender and purchase outcome, plus supporting breakdowns by region, education, and marital status.
- **Interactive dashboard** — a single-page dashboard combining bar, line, and pie charts, with three linked slicers (Marital Status, Region, Education) so a viewer can filter every chart at once without touching a single formula.

Why this project

Most sample dashboards just decorate data. This one is built to be interrogated — the slicers let you slice the customer base by segment and watch the averages shift in real time, which is closer to how a stakeholder actually uses a report in a meeting.

Skills demonstrated

- Data cleaning and validation in Excel
- Nested conditional logic (IF) for feature engineering
- PivotTables and PivotCharts
- Cross-filtering with slicers connected to multiple pivot tables
- Dashboard layout and design for non-technical end users

A note on the data: this dataset is a well-known public bike-buyer dataset used for BI/analytics practice. Nothing in it is proprietary or personal.

Key Insights & Business Recommendations

- **Women convert better than men** (41.1% vs. 32.4%) — marketing spend aimed at male customers may be underperforming and worth re-evaluating.
- **Single customers convert far more than married customers** (54.1% vs. 42.9%) — campaigns built around individual lifestyle/independence may resonate more than family-oriented messaging.
- **Pacific region outperforms North America** (58.9% vs. 43.3% conversion) — regional demand differs sharply, so a one-size-fits-all sales strategy is leaving conversions on the table in underperforming regions.
- **Education level matters** — customers with only partial high school convert far worse (26.3%) than every other education group, suggesting this segment may need a different offer, price point, or messaging entirely.

Business takeaway: the highest-converting customer profile is a single woman in the Pacific region — that's the segment to prioritize in targeted campaigns, while North America and lower-education segments need a revised approach rather than the current one.

Possible next steps

- Break out commute distance and cars-owned as their own filterable segments, since they're currently in the raw data but not surfaced on the dashboard
- Test whether income or age is the stronger predictor of purchase by comparing conversion rates across both dimensions side by side